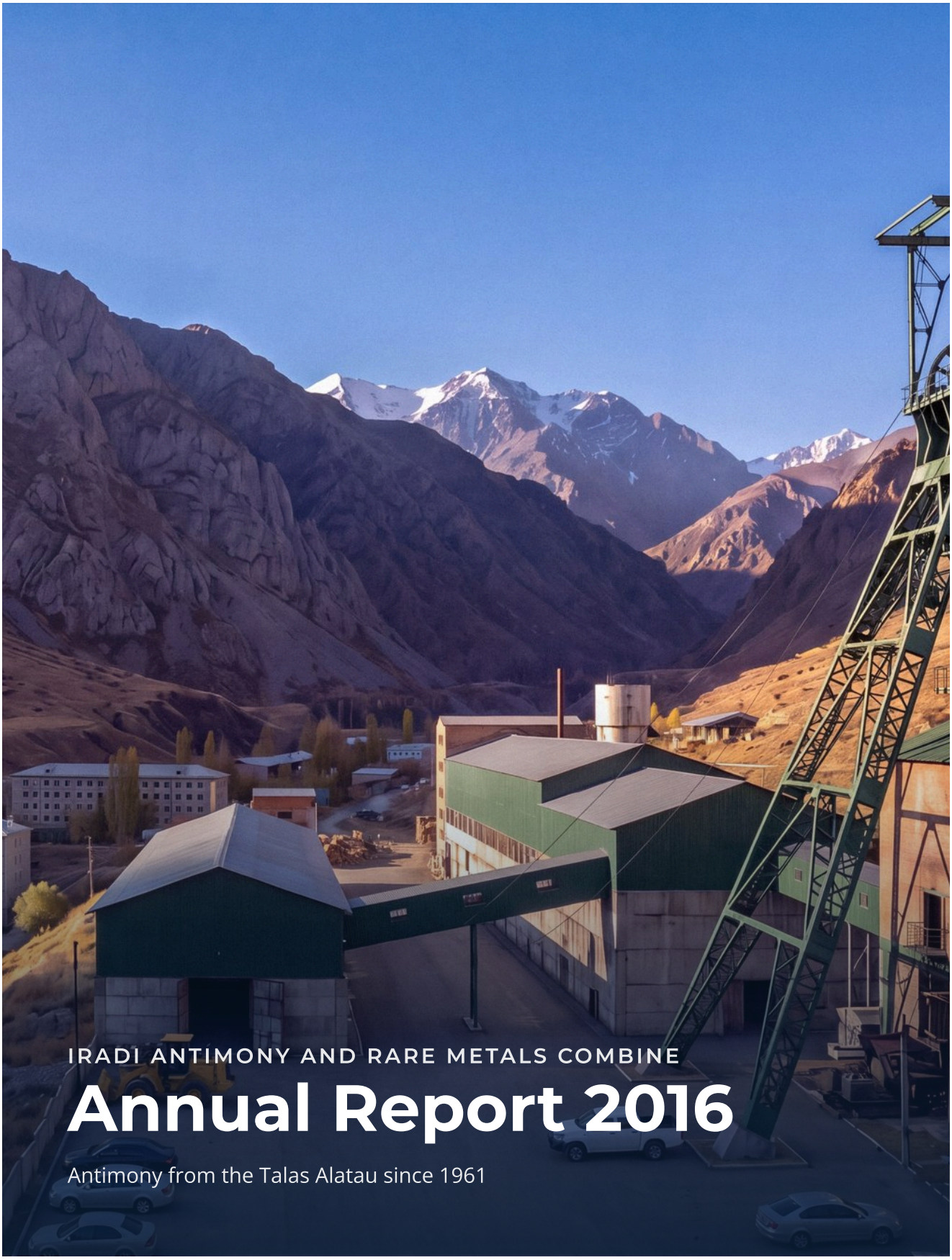
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IRADI ANTIMONY AND RARE METALS COMBINE



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# Annual Report 2016

Antimony from the Talas Alatau since 1961

Talas, 2017

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## 1. The year at a glance

**12,500 t**
antimony mined (contained)

**3,400 t**
antimony metal produced at Chatkal

**US$70 m**
revenue

**0**
fatal accidents

### Key figures

| INDICATOR | UNIT | 2014 | 2015 | 2016 |
|---|---|---|---|---|
| Antimony mined (contained in ore) | t | 13,400 | 12,800 | 12,500 |
| Antimony metal (Chatkal) | t | 3,550 | 3,450 | 3,400 |
| Antimony trioxide (Chatkal) | t | 2,200 | 2,150 | 2,200 |
| Antimony sold in concentrate (contained) | t | 8,000 | 7,600 | 7,300 |
| Revenue | US$ m | 104 | 78 | 70 |
| Royalties and export duty paid | US$ m | 12 | 9 | 8 |
| Capital expenditure | US$ m | 10 | 6 | 9 |
| Employees at year end | persons | 4,800 | 4,690 | 4,610 |

> Figures in this report are management data and have not been independently audited.

# 2. Letter from the General Director

Dear shareholders,

In 2016 the Combine mined 12,500 tonnes of contained antimony. Antimony prices stayed low for most of the year, and revenue was US$70 million. Prices began to recover in the last quarter.

The sulphur dioxide scrubbers and electrostatic precipitators at the Chatkal complex entered service in 2016. They were designed under the direction of the chief engineer, Artem Kozlov, and they reduce emissions of sulphur dioxide and dust from the roasters.

No fatal accident occurred at any of our operations in 2016.

The Supervisory Board has approved a gas-capture unit at Chatkal that will turn sulphur dioxide into sulphuric acid. Construction will begin in 2017.

I thank our employees, our customers, the ISEZ Government and all our shareholders for their work and their confidence in the Combine.

**Almaz Toktosunov**
General Director
Talas, June 2017



The Rudnik settlement beside the Sary-Bulak mine.

# 3. Operations

## Sary-Bulak mine

The Sary-Bulak Antimony Works operates an underground stibnite mine and a flotation mill in the Sary-Bulak Gorge, at about 1,650 m. Ore is reached through Shaft No. 1 "Pioneer", Shaft No. 2 "Severny" and the Central incline, to a depth of about 750 m. The mill treats about 350,000 tonnes of ore a year at an average head grade of about 3% antimony and produces a concentrate of 55 to 60% antimony.

## Satellite mines

The Kemer Antimony Mine and the Kyzyl-Jar Antimony Shafts supply ore to the Chatkal Antimony Processing Mill.

## Chatkal complex

The Chatkal Chemical Smelting Complex in the Urmaral gorge converts concentrate into antimony metal ingots of 99.65% Sb and high-purity antimony trioxide. Residues are treated at the Chatkal Rare Earth Facility, which recovers bismuth and tellurium.

## Workforce

At the end of 2016 the Combine employed 4,610 people: about 2,300 at Sary-Bulak, about 2,200 at Chatkal and the rest at the head office and services.



The Chatkal smelter above the processing mill, Urmaral gorge.

### Antimony mined by quarter, 2016 (t contained)

| QUARTER | Q1 | Q2 | Q3 | Q4 | YEAR |
|---|---|---|---|---|---|
| Antimony mined | 3,000 | 3,120 | 3,190 | 3,190 | 12,500 |

## 4. Sales and markets

The Combine sells antimony metal, antimony trioxide, antimony concentrate, bismuth and tellurium to customers in China, South Korea, Japan and Europe, mostly under long-term contracts. Contracts are concluded in US dollars. Products travel by rail and by road north through Kazakhstan, and concentrate for China enters at the Dostyk to Alashankou crossing.

Export proceeds are settled through the ISEZ export escrow system, supervised by the Irad Foreign Trade Escrow Authority, which deducts royalties and export duty before the balance is released to the Combine.



A concentrate train on the Sary-Bulak spur.

## Investment programme

Capital expenditure in 2016 was US$9 million. The main items were the completion of the sulphur dioxide scrubbers and electrostatic precipitators at Chatkal.

# 5. Financial indicators and payments to the ISEZ budget

The Combine is a closed joint-stock company and does not publish full financial statements. The indicators below are taken from management accounts.

### Financial indicators, US$ m

| INDICATOR | 2014 | 2015 | 2016 |
|---|---|---|---|
| Revenue | 104 | 78 | 70 |
| Capital expenditure | 10 | 6 | 9 |

### Payments to the ISEZ budget, US$ m

| PAYMENT | 2014 | 2015 | 2016 |
|---|---|---|---|
| Royalties and antimony export duty | 12 | 9 | 8 |
| Profit tax | 2 | 0 | 0 |
| Payroll taxes and social contributions | 7 | 8 | 8 |
| Land, water and environmental fees | 1 | 1 | 1 |
| **Total** | **22** | **18** | **17** |

Royalties are set under the Hydrocarbon and Mining Code of 1995. Under formulas adopted by the Territorial Legislature in the 1990s, antimony revenue is divided between infrastructure in Talas and rural schools.

> Figures are management data and have not been independently audited.



Antimony ingots on steel-strapped pallets, Chatkal.

# 6. Health, safety and the environment

### Safety indicators

| INDICATOR | 2014 | 2015 | 2016 |
|---|---|---|---|
| Fatal accidents | 0 | 1 | 0 |
| Lost-time injuries | 60 | 58 | 55 |
| Lost-time injury frequency rate (per 1 million hours worked) | 4.4 | 4.3 | 4.2 |

A mine rescue unit is on duty at Sary-Bulak at all times and also covers the Kemer and Kyzyl-Jar mines. Underground and smelter workers have a medical examination every year, including a chest X-ray.

No fatal accident occurred in 2016.

### Air

Sulphur dioxide scrubbers and electrostatic precipitators have been in operation at Chatkal since 2016. A gas-capture unit to turn sulphur dioxide into sulphuric acid is under construction. Emissions meet ISEZ standards.

### Water

Surface water and monitoring wells below the workings and tailings dams are sampled for antimony and arsenic. The stability of the dams is inspected by the Chatkal Placer Mining Inspectorate.

### Tailings

Tailings from current production are stored at Sary-Bulak and Chatkal. Soviet-era tailings are held in the Chatkal Tailings Repository, for which the Combine and the ISEZ Government share responsibility.

### Community

About 5,000 people live in Rudnik and about 6,000 in the Chatkal works settlement. The technical lyceum at Chatkal trains metallurgical technicians.



The house of culture in Rudnik.

# 7. Governance

The Combine is a closed joint-stock company registered in Talas. Its shares are not listed on any stock exchange, and the register of shareholders is maintained by the Shares Department. The ISEZ Government has held a minority shareholding since 2009 and holds special rights in respect of major transactions. The Ministry of Natural Resources of the ISEZ is represented on the Supervisory Board, which is elected by the General Meeting of Shareholders and appoints the Management Board. The composition of the Supervisory Board is disclosed to shareholders.

### Management Board

| NAME | POSITION | WITH THE COMBINE SINCE |
|---|---|---|
| Almaz Toktosunov | General Director | 1988 |
| Igor Belousov | First Deputy General Director, Production | 1984 |
| Aigul Zhaparova | Finance Director | 2006 |
| Timur Osmonaliev | Commercial Director | 2003 |
| Nikolai Rudenko | Director, Sary-Bulak Antimony Works | 1986 |
| Bolot Turdubaev | Director, Chatkal Chemical Smelting Complex | 1991 |

The Annual General Meeting of Shareholders is held in June in Talas.

**Important information.** Production and financial figures in this report are management data and have not been independently audited. Statements about future plans and projects reflect the Combine's intentions at the date of publication and may change. Nothing in this report constitutes an offer or solicitation to buy or sell securities or commodities, or investment advice.

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